

Week 2: DAY-4 Hands-On

Level-1 Problem 1: Responsive Personal Portfolio Homepage

Scenario:

You are designing a responsive personal portfolio homepage for a fresher web developer who wants to showcase profile information, skills, and contact details.

Requirements

- Integrate Bootstrap 5 using CDN.
- Create a responsive layout using container, rows, and columns.
- Use Bootstrap grid system with appropriate breakpoints.
- Include components such as navbar, cards (for skills), and buttons.
- Use utility classes for spacing, alignment, and colors.

Technical Constraints

- Must use Bootstrap 5 CDN (no local files).
- No custom CSS except minimal styling if required.
- Use grid system (row and col classes) properly.
- Ensure responsiveness across mobile, tablet, and desktop.

Learning Outcome

- Understand Bootstrap setup using CDN.
- Learn responsive grid system and breakpoints.
- Implement basic Bootstrap components.
- Apply utility classes effectively.

CODE:

```
<!DOCTYPE html>
<html lang="en">

<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Personal Portfolio</title>
  <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/css/bootstrap.min.css"
rel="stylesheet">
</head>

<body class="bg-light">

  <nav class="navbar navbar-expand-lg navbar-dark bg-dark">
    <div class="container">
      <a class="navbar-brand" href="#">My Portfolio</a>
      <button class="navbar-toggler" type="button" data-bs-toggle="collapse" data-bs-
target="#navbarNav">
        <span class="navbar-toggler-icon"></span>
      </button>
      <div class="collapse navbar-collapse" id="navbarNav">
        <ul class="navbar-nav ms-auto">
          <li class="nav-item"><a class="nav-link" href="#about">About</a></li>
          <li class="nav-item"><a class="nav-link" href="#skills">Skills</a></li>
          <li class="nav-item"><a class="nav-link" href="#contact">Contact</a></li>
        </ul>
      </div>
    </div>
  </nav>

  <header id="about" class="container text-center py-5 mt-5 bg-white shadow-sm rounded">
    <h1 class="display-4 fw-bold text-primary">Hello, I'm Keerthi</h1>
    <p class="lead text-muted">A passionate Fresher Web Developer building modern,
responsive websites.</p>
    <a href="#contact" class="btn btn-primary btn-lg mt-3">Hire Me</a>
    <a href="#skills" class="btn btn-outline-secondary btn-lg mt-3 ms-2">View Skills</a>
  </header>

  <section id="skills" class="container py-5">
    <h2 class="text-center mb-4">My Skills</h2>
    <div class="row g-4">
      <div class="col-12 col-md-4">
        <div class="card h-100 shadow-sm border-0 text-center">
          <div class="card-body">
            <h5 class="card-title text-success">Frontend</h5>
            <p class="card-text">HTML5, CSS3, JavaScript (ES6+), and Bootstrap
5.</p>
          </div>
        </div>
      </div>
    </div>
  </section>
</body>
</html>
```

```

        </div>
    </div>
    <div class="col-12 col-md-4">
        <div class="card h-100 shadow-sm border-0 text-center">
            <div class="card-body">
                <h5 class="card-title text-info">Backend</h5>
                <p class="card-text">Basic understanding of Node.js and Express
server routing.</p>
            </div>
        </div>
    </div>
    <div class="col-12 col-md-4">
        <div class="card h-100 shadow-sm border-0 text-center">
            <div class="card-body">
                <h5 class="card-title text-warning">Tools</h5>
                <p class="card-text">Git, GitHub, VS Code, and responsive design
principles.</p>
            </div>
        </div>
    </div>
</div>
</section>

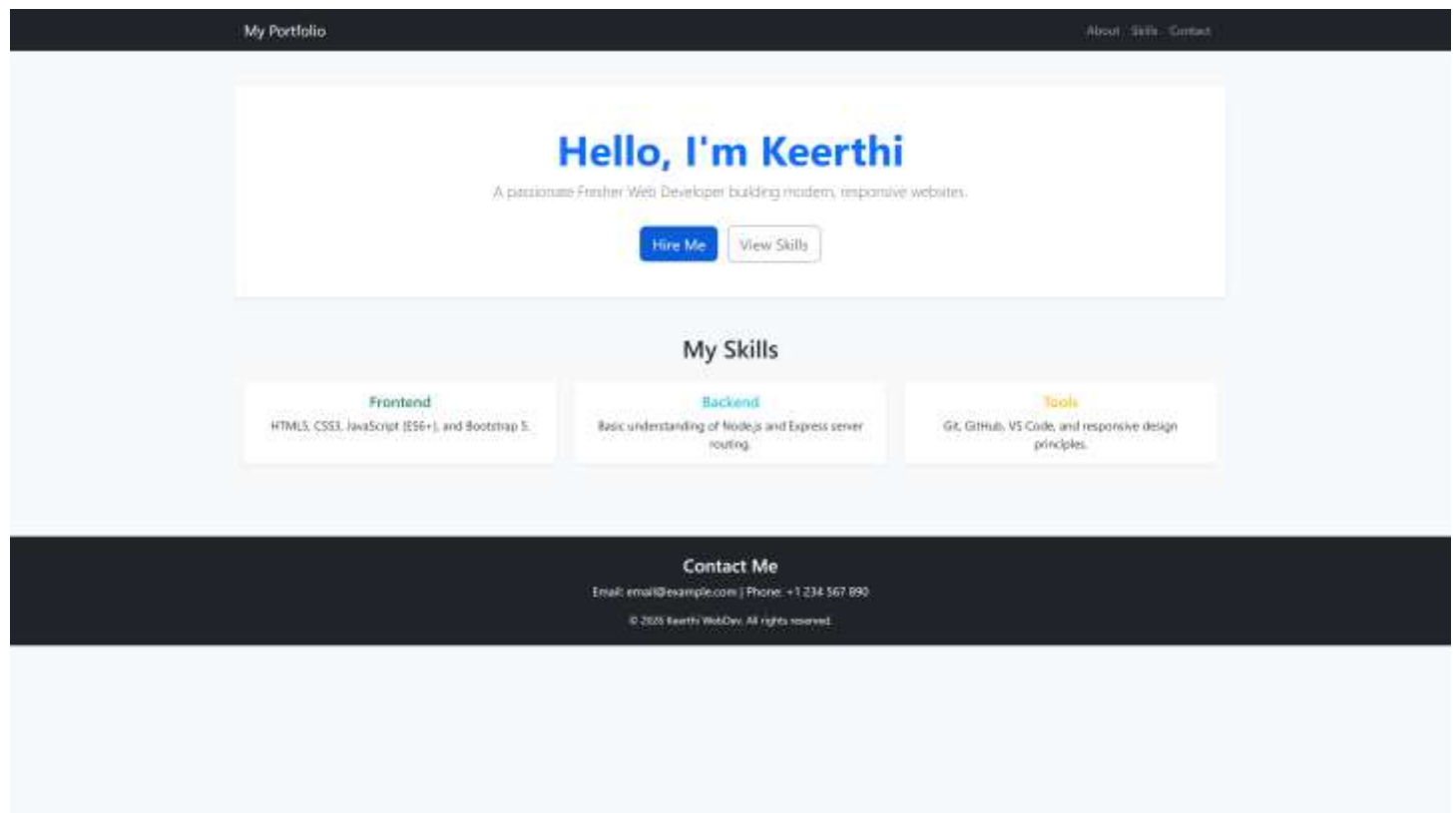
<footer id="contact" class="bg-dark text-white text-center py-4 mt-5">
    <div class="container">
        <h4>Contact Me</h4>
        <p>Email: email@example.com | Phone: +1 234 567 890</p>
        <p class="mb-0 small">&copy; 2026 Keerthi WebDev. All rights reserved.</p>
    </div>
</footer>

<script
src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/js/bootstrap.bundle.min.js"></script>
</body>

</html>

```

OUTPUT:



Level-1 Problem 2: Product Listing Page Using Bootstrap

Scenario:

Create a responsive product listing page for an online electronics store displaying multiple products in card format.

📌 Requirements

- Use Bootstrap grid to display products in 1 column (mobile), 2 columns (tablet), and 4 columns (desktop).
- Each product should be displayed using Bootstrap Card component.
- Include buttons such as "Add to Cart" and "View Details".

- Use spacing utilities and background color utilities.

Technical Constraints

- Must use Bootstrap CDN.
- Use built-in Bootstrap classes only.
- Proper use of container, row, and col classes.

Expectations:

- Fully responsive layout.
- Equal card height alignment.
- Proper spacing using margin and padding utilities.

Learning Outcome

- Master grid layouts with multiple breakpoints.
- Use Bootstrap cards and buttons.
- Apply spacing and color utility classes.

CODE:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Electronics Store</title>
  <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/css/bootstrap.min.css"
rel="stylesheet">
</head>
<body class="bg-light">

  <div class="container py-5">
    <h1 class="text-center mb-5">Featured Electronics</h1>

    <div class="row g-4">

      <div class="col-12 col-md-6 col-lg-3">
```

```

        <div class="card h-100 shadow-sm border-0">
            
            <div class="card-body d-flex flex-column">
                <h5 class="card-title">Wireless Headphones</h5>
                <p class="card-text text-muted mb-4">Noise-cancelling over-ear
headphones.</p>
                <h4 class="text-primary mb-3">$199.99</h4>
                <div class="mt-auto d-grid gap-2">
                    <button class="btn btn-primary">Add to Cart</button>
                    <button class="btn btn-outline-secondary">View Details</button>
                </div>
            </div>
        </div>
    </div>

    <div class="col-12 col-md-6 col-lg-3">
        <div class="card h-100 shadow-sm border-0">
            
            <div class="card-body d-flex flex-column">
                <h5 class="card-title">Smart Watch</h5>
                <p class="card-text text-muted mb-4">Fitness tracker with heart rate
monitor.</p>
                <h4 class="text-primary mb-3">$149.00</h4>
                <div class="mt-auto d-grid gap-2">
                    <button class="btn btn-primary">Add to Cart</button>
                    <button class="btn btn-outline-secondary">View Details</button>
                </div>
            </div>
        </div>
    </div>

    <div class="col-12 col-md-6 col-lg-3">
        <div class="card h-100 shadow-sm border-0">
            
            <div class="card-body d-flex flex-column">
                <h5 class="card-title">4K Action Camera</h5>
                <p class="card-text text-muted mb-4">Waterproof camera for extreme
sports.</p>
                <h4 class="text-primary mb-3">$299.50</h4>
                <div class="mt-auto d-grid gap-2">
                    <button class="btn btn-primary">Add to Cart</button>
                    <button class="btn btn-outline-secondary">View Details</button>
                </div>
            </div>
        </div>
    </div>
</div>

```

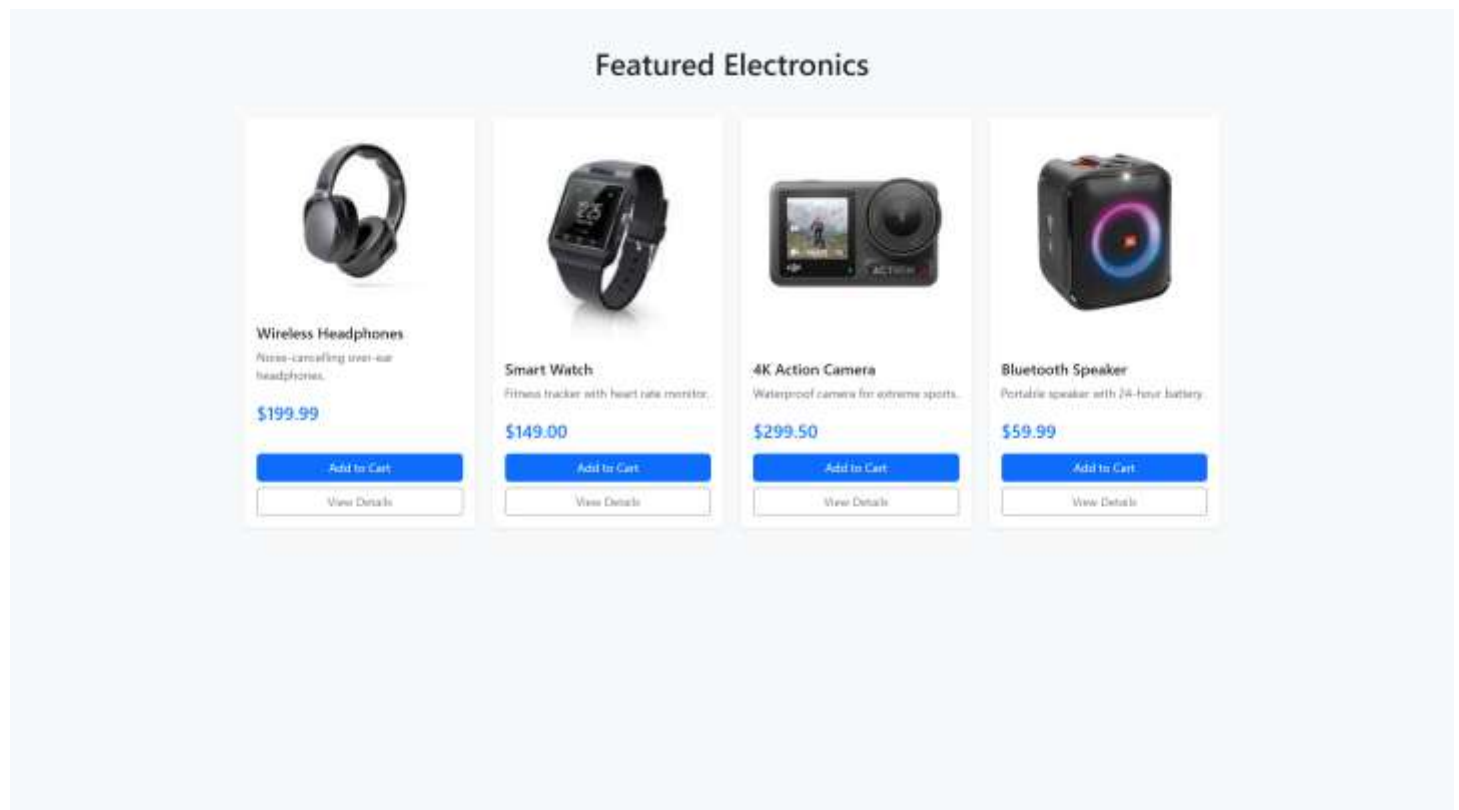
```

<div class="col-12 col-md-6 col-lg-3">
  <div class="card h-100 shadow-sm border-0">
    
    <div class="card-body d-flex flex-column">
      <h5 class="card-title">Bluetooth Speaker</h5>
      <p class="card-text text-muted mb-4">Portable speaker with 24-hour
battery.</p>
      <h4 class="text-primary mb-3">$59.99</h4>
      <div class="mt-auto d-grid gap-2">
        <button class="btn btn-primary">Add to Cart</button>
        <button class="btn btn-outline-secondary">View Details</button>
      </div>
    </div>
  </div>
</div>

<script
src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/js/bootstrap.bundle.min.js"></script>
</body>
</html>

```

OUTPUT:



Level-2 Problem 3: College Event Management Dashboard

Scenario:

Design a responsive dashboard for managing college events including event list, add event form, and navigation.

Requirements

- Use Bootstrap navbar with collapse functionality.
- Implement responsive grid layout for dashboard sections.
- Use form components for event registration.
- Implement modal for adding new events.
- Use Bootstrap utilities for alignment and spacing.

Technical Constraints

- Use Bootstrap 5 CDN.
- No external UI frameworks.
- Proper breakpoint usage (sm, md, lg, xl).
- Modal must work using Bootstrap JS bundle.

Expectations:

- Dashboard layout adapts across devices.
- Modal opens and closes correctly.
- Form fields use proper Bootstrap validation styling.

Learning Outcome

- Implement advanced components like modals.
- Create structured responsive dashboards.
- Apply responsive behaviors effectively.

CODE:

```
<!DOCTYPE html>
<html lang="en">

<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Event Dashboard</title>
  <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/css/bootstrap.min.css"
rel="stylesheet">
</head>

<body>

  <nav class="navbar navbar-expand-md navbar-dark bg-primary shadow-sm">
    <div class="container-fluid">
      <a class="navbar-brand fw-bold" href="#">UniEvents</a>
      <button class="navbar-toggler" type="button" data-bs-toggle="collapse" data-bs-
target="#dashboardNav">
        <span class="navbar-toggler-icon"></span>
      </button>
      <div class="collapse navbar-collapse" id="dashboardNav">
        <ul class="navbar-nav me-auto">
          <li class="nav-item"><a class="nav-link active"
href="#">Dashboard</a></li>
          <li class="nav-item"><a class="nav-link" href="#">Reports</a></li>
        </ul>
        <button class="btn btn-light text-primary fw-bold" data-bs-toggle="modal"
data-bs-target="#addEventModal">+ Add Event</button>
      </div>
    </div>
  </nav>

  <div class="container py-4">
    <div class="row g-4">

      <div class="col-12 col-lg-3">
        <div class="card shadow-sm border-0 mb-3">
          <div class="card-body text-center">
            <h6 class="text-muted">Total Events</h6>
            <h2 class="text-primary mb-0">24</h2>
          </div>
        </div>
        <div class="list-group shadow-sm">
          <a href="#" class="list-group-item list-group-item-action
active">Upcoming Events</a>
          <a href="#" class="list-group-item list-group-item-action">Past
Events</a>
```

```

                <a href="#" class="list-group-item list-group-item-action">Pending
Approvals</a>
            </div>
        </div>

        <div class="col-12 col-lg-9">
            <div class="card shadow-sm border-0">
                <div class="card-header bg-white d-flex justify-content-between align-
items-center py-3">
                    <h5 class="mb-0">Upcoming Events</h5>
                </div>
                <div class="table-responsive">
                    <table class="table table-hover mb-0 align-middle">
                        <thead class="table-light">
                            <tr>
                                <th>Event Name</th>
                                <th>Date</th>
                                <th>Location</th>
                                <th>Status</th>
                            </tr>
                        </thead>
                        <tbody>
                            <tr>
                                <td>Tech Symposium 2026</td>
                                <td>March 15, 2026</td>
                                <td>Main Auditorium</td>
                                <td><span class="badge bg-success">Confirmed</span></td>
                            </tr>
                            <tr>
                                <td>Annual Sports Meet</td>
                                <td>April 02, 2026</td>
                                <td>College Ground</td>
                                <td><span class="badge bg-warning text-
dark">Pending</span></td>
                            </tr>
                        </tbody>
                    </table>
                </div>
            </div>
        </div>

        <div class="modal fade" id="addEventModal" tabindex="-1" aria-
labelledby="addEventModalLabel" aria-hidden="true">
            <div class="modal-dialog">
                <div class="modal-content">
                    <div class="modal-header bg-light">
                        <h5 class="modal-title" id="addEventModalLabel">Register New Event</h5>

```

```

        <button type="button" class="btn-close" data-bs-dismiss="modal" aria-label="Close"></button>
      </div>
      <div class="modal-body">
        <form class="was-validated">
          <div class="mb-3">
            <label for="eventName" class="form-label">Event Name</label>
            <input type="text" class="form-control" id="eventName" required>
            <div class="invalid-feedback">Please provide an event name.</div>
          </div>
          <div class="mb-3">
            <label for="eventDate" class="form-label">Date</label>
            <input type="date" class="form-control" id="eventDate" required>
          </div>
          <div class="mb-3">
            <label for="eventLocation" class="form-label">Location</label>
            <select class="form-select" id="eventLocation" required>
              <option value="">Choose...</option>
              <option value="1">Main Auditorium</option>
              <option value="2">Seminar Hall B</option>
              <option value="3">College Ground</option>
            </select>
            <div class="invalid-feedback">Please select a location.</div>
          </div>
          <div class="d-grid">
            <button type="submit" class="btn btn-primary">Save Event</button>
          </div>
        </form>
      </div>
    </div>
  </div>
</div>

<script
src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/js/bootstrap.bundle.min.js"></script>
</body>

</html>

```

OUTPUT:

UniEvents

Dashboard Reports

+ Add Event

Total Events

24

Upcoming Events

Past Events

Pending Approvals

Upcoming Events

Event Name	Date	Location	Status
Tech Symposium 2026	March 15, 2026	Main Auditorium	Confirmed
Annual Sports Meet	April 02, 2026	College Ground	Pending

Level-2 Problem 4: E-Commerce Landing Page with Advanced Features

Scenario:

Build a professional e-commerce landing page with hero section, product carousel, offcanvas menu, and themed design.

Requirements

- Use responsive navbar with offcanvas menu.
- Implement Bootstrap Carousel for featured products.
- Create multiple sections using grid system.
- Use forms for newsletter subscription.
- Apply Bootstrap utilities for display control and alignment.
- Customize theme using Bootstrap variables or custom overrides.

Technical Constraints

- Bootstrap 5 CDN required.
- Proper responsive behavior implementation.
- Offcanvas and Carousel must function correctly.
- Maintain consistent color theme.

Expectations:

- Professional UI design.
- Smooth functioning carousel and offcanvas menu.
- Responsive layout for all devices.

Learning Outcome

- Implement advanced Bootstrap components.
- Understand responsive behavior and display utilities.
- Apply custom theming concepts.

CODE:

```
<!DOCTYPE html>
<html lang="en">

<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Modern E-Commerce</title>
  <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/css/bootstrap.min.css"
rel="stylesheet">
  <style>
    /* Minimal Custom Theming / Overrides */
    :root {
      --bs-primary: #ff6b6b;
      /* Themed Primary Color */
    }

    .btn-primary {
      background-color: var(--bs-primary);
      border-color: var(--bs-primary);
    }

    .btn-primary:hover {
      background-color: #ff5252;
      border-color: #ff5252;
    }

    .text-theme {
      color: var(--bs-primary);
    }
  </style>
</head>

<body>

  <nav class="navbar navbar-expand-lg navbar-light bg-white sticky-top shadow-sm py-3">
    <div class="container">
      <a class="navbar-brand fw-bold fs-3 text-theme" href="#">Shopify</a>
      <button class="navbar-toggler border-0" type="button" data-bs-toggle="offcanvas"
        data-bs-target="#offcanvasNavbar">
        <span class="navbar-toggler-icon"></span>
      </button>

      <div class="offcanvas offcanvas-end" tabindex="-1" id="offcanvasNavbar"
        aria-labelledby="offcanvasNavbarLabel">
        <div class="offcanvas-header bg-light">
          <h5 class="offcanvas-title" id="offcanvasNavbarLabel">Menu</h5>
          <button type="button" class="btn-close" data-bs-dismiss="offcanvas" aria-
label="Close"></button>
        </div>
      </div>
    </div>
  </nav>
</body>
</html>
```

```

        </div>
        <div class="offcanvas-body">
            <ul class="navbar-nav justify-content-center flex-grow-1 pe-3">
                <li class="nav-item"><a class="nav-link active"
href="#">Home</a></li>
                <li class="nav-item"><a class="nav-link" href="#">Men</a></li>
                <li class="nav-item"><a class="nav-link" href="#">Women</a></li>
                <li class="nav-item"><a class="nav-link" href="#">Sale</a></li>
            </ul>
            <div class="d-flex align-items-center">
                <button class="btn btn-outline-dark me-2">Log In</button>
                <button class="btn btn-primary">Cart (0)</button>
            </div>
        </div>
    </div>
</div>
</nav>

<header id="heroCarousel" class="carousel slide" data-bs-ride="carousel">
    <div class="carousel-indicators">
        <button type="button" data-bs-target="#heroCarousel" data-bs-slide-to="0"
class="active"></button>
        <button type="button" data-bs-target="#heroCarousel" data-bs-slide-
to="1"></button>
    </div>
    <div class="carousel-inner">
        <div class="carousel-item active" style="height: 60vh; background-color:
#f8f9fa;">
            <div class="container h-100 d-flex align-items-center justify-content-center
text-center">
                <div>
                    <h1 class="display-3 fw-bold">Summer Collection</h1>
                    <p class="lead mb-4">Up to 50% off on selected items.</p>
                    <a href="#" class="btn btn-primary btn-lg px-5 rounded-pill">Shop
Now</a>
                </div>
            </div>
        </div>
        <div class="carousel-item" style="height: 60vh; background-color: #e9ecef;">
            <div class="container h-100 d-flex align-items-center justify-content-center
text-center">
                <div>
                    <h1 class="display-3 fw-bold">New Arrivals</h1>
                    <p class="lead mb-4">Discover the latest trends in fashion.</p>
                    <a href="#" class="btn btn-dark btn-lg px-5 rounded-pill">Explore</a>
                </div>
            </div>
        </div>
    </div>
</div>

```

```

        <button class="carousel-control-prev" type="button" data-bs-target="#heroCarousel"
data-bs-slide="prev">
            <span class="carousel-control-prev-icon text-dark" aria-hidden="true"></span>
        </button>
        <button class="carousel-control-next" type="button" data-bs-target="#heroCarousel"
data-bs-slide="next">
            <span class="carousel-control-next-icon" aria-hidden="true"></span>
        </button>
    </header>

    <section class="container py-5 mt-4">
        <h2 class="text-center fw-bold mb-5">Trending Now</h2>
        <div class="row g-4">
            <div class="col-12 col-md-4">
                <div class="card border-0 shadow-sm">
                    
                    <div class="card-body text-center">
                        <h5 class="card-title">Classic T-Shirt</h5>
                        <p class="text-theme fw-bold">$25.00</p>
                    </div>
                </div>
            </div>
            <div class="col-12 col-md-4">
                <div class="card border-0 shadow-sm">
                    
                    <div class="card-body text-center">
                        <h5 class="card-title">Denim Jacket</h5>
                        <p class="text-theme fw-bold">$89.00</p>
                    </div>
                </div>
            </div>
            <div class="col-12 col-md-4">
                <div class="card border-0 shadow-sm">
                    
                    <div class="card-body text-center">
                        <h5 class="card-title">Running Sneakers</h5>
                        <p class="text-theme fw-bold">$120.00</p>
                    </div>
                </div>
            </div>
        </div>
    </section>

    <section class="bg-dark text-white py-5 text-center">
        <div class="container py-4">
            <h3 class="mb-3">Subscribe to our Newsletter</h3>

```

```

    <p class="mb-4 text-light">Get the latest updates on new products and upcoming
sales.</p>
    <div class="row justify-content-center">
      <div class="col-12 col-md-6">
        <form class="input-group input-group-lg">
          <input type="email" class="form-control border-0" placeholder="Enter
your email address"
              required>
          <button class="btn btn-primary px-4" type="submit">Subscribe</button>
        </form>
      </div>
    </div>
  </div>
</section>

<script
src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/js/bootstrap.bundle.min.js"></script>
</body>

</html>

```

OUTPUT:

